SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMKUR-3
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Problems on Trees

Faculty: Smt. Kavitha M **Class:** III Sem

1. Find the following for the tree structure given:

a) Height of the tree: **3**

b) Leaves at level 2: **E, I**

c) Ancestors of node F: **A, C**

d) Descendants of node B: **D, E, G**

e) Indegree of node B: **1**

f) Siblings at level2: **D & E, F & I**

g) Degree of node C: **3**

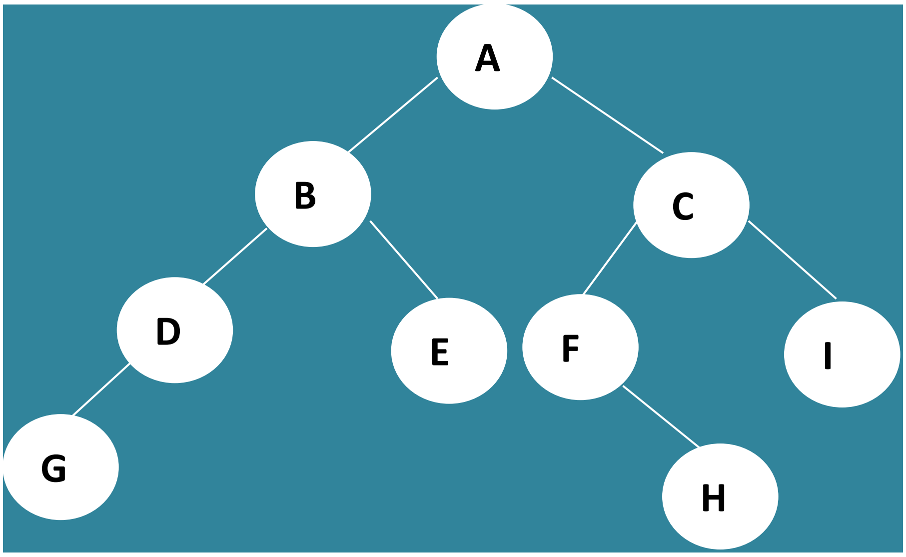
h) Level of node I : **2**

i) Traverse the tree in inorder, preorder and postorder

Inorder: **GDBEAFHCI**

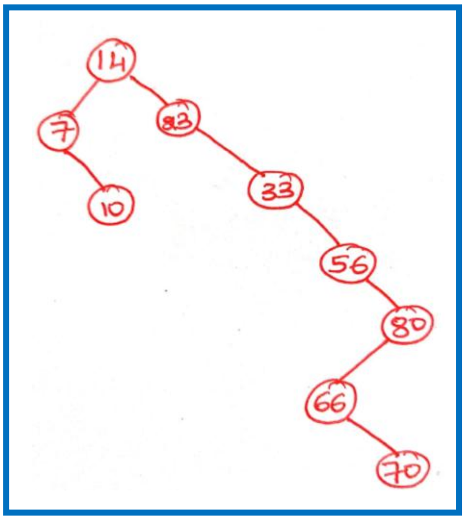
Preorder: **ABDGECFHI**

Postorder: **GDEBHFICA**

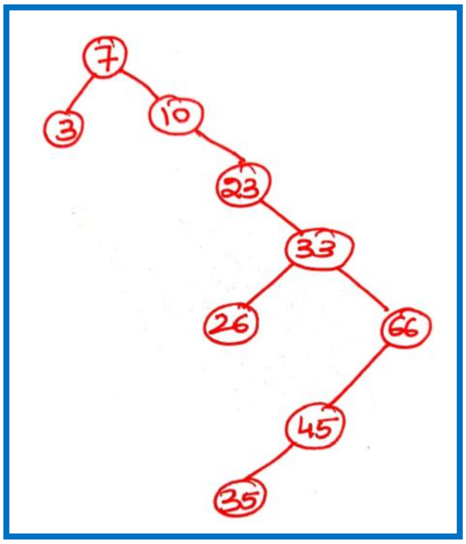


2. **Construct a Binary Search tree using the following data entered as a sequential set.**

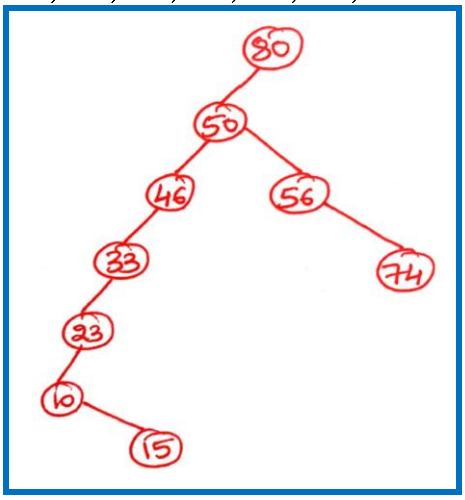
a) 14, 23, 7, 10, 33, 56, 80, 66, 70



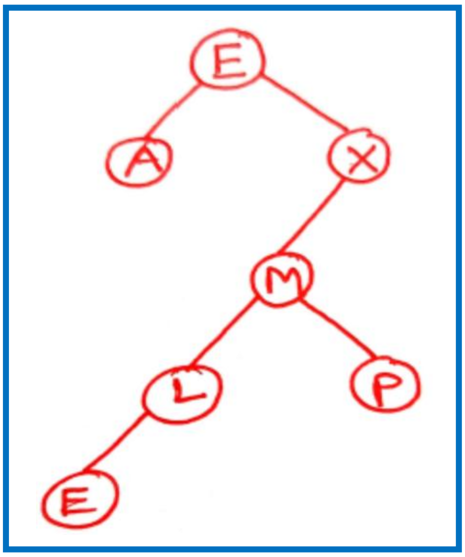
b) 7, 10, 3, 23, 33, 26, 66, 45, 35



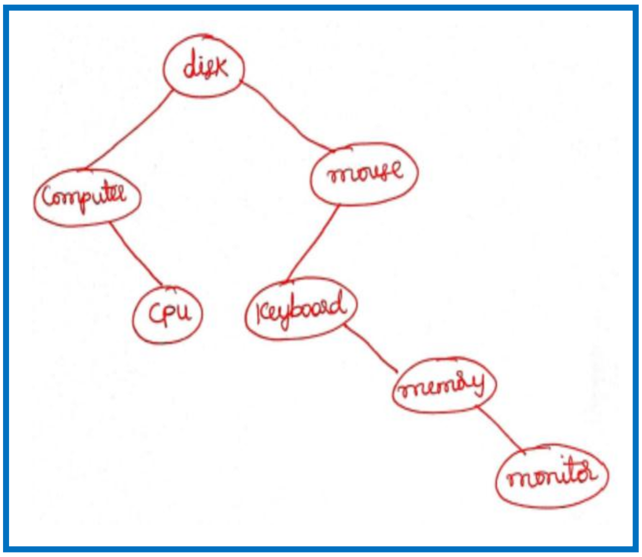
c) 80, 50, 46, 56, 33, 23, 74, 10, 15



d) E, X, A, M, P, L, E



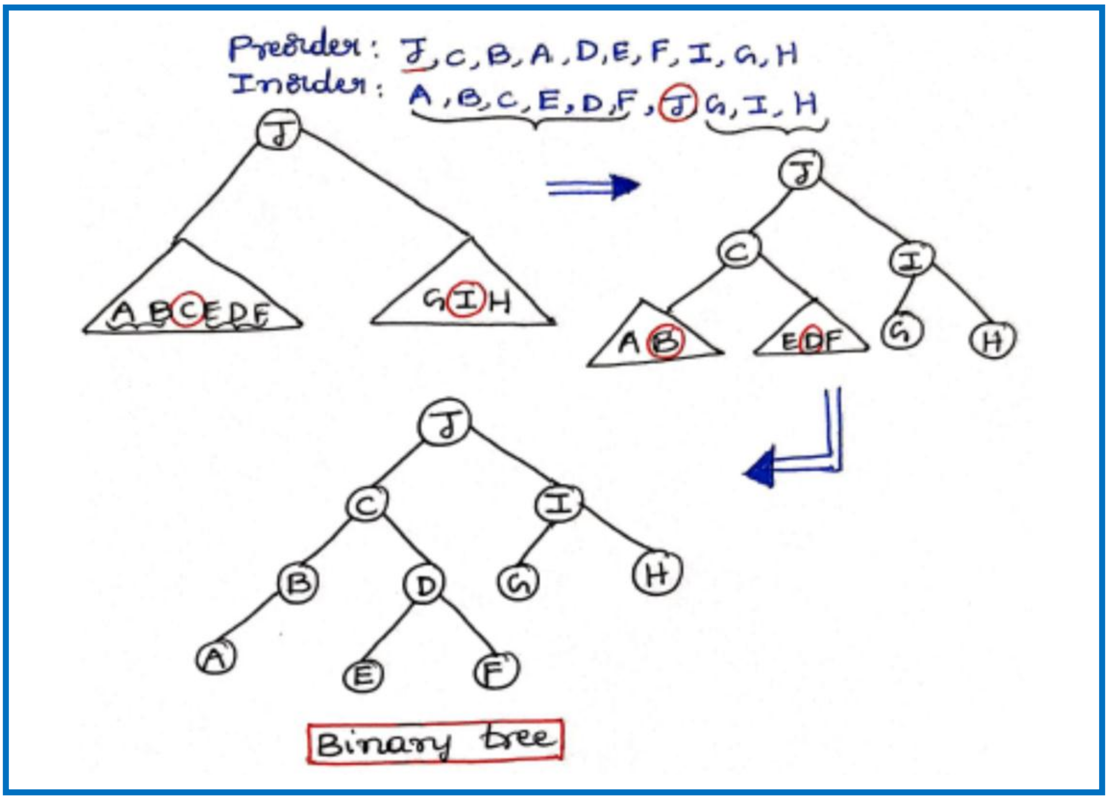
e) “disk”, “computer”, “mouse”, “keyboard”, “memory”, “cpu”, “monitor”



3. Construct a Binary tree for the following data:

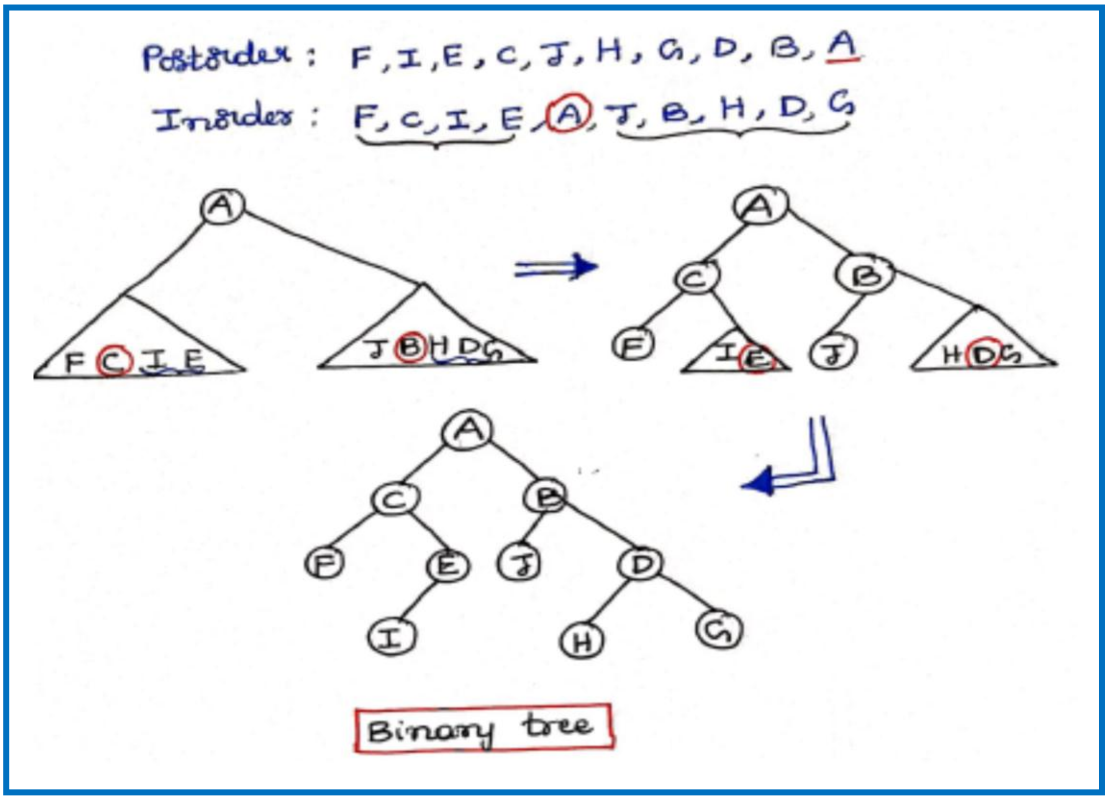
a) **Preorder:** J, C, B, A, D, E, F, I, G, H

Inorder: A, B, C, E, D, F, J, G, I, H



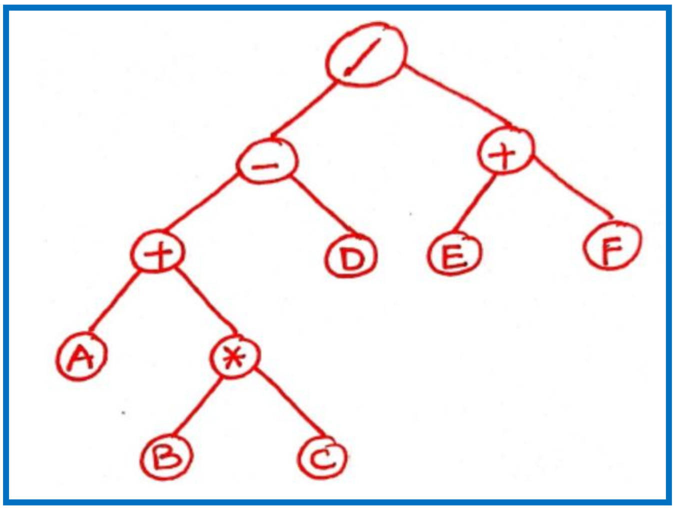
b) **Postorder:** F, I, E, C, J, H, G, D, B, A

Inorder: F, C , I, E, A, J, B, H, D, G

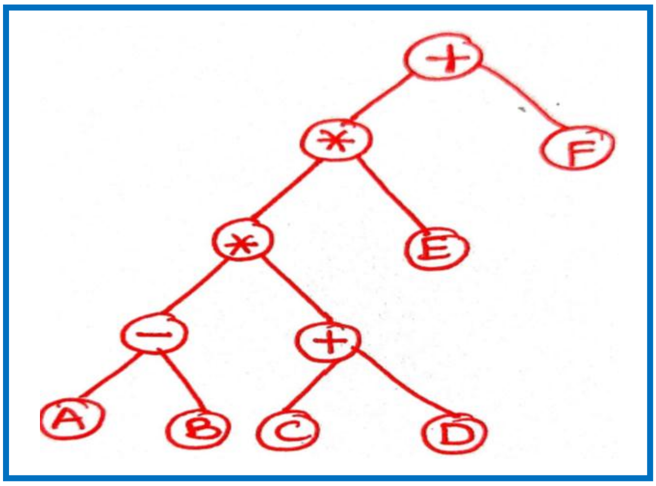


4. Construct the expression tree for the following:

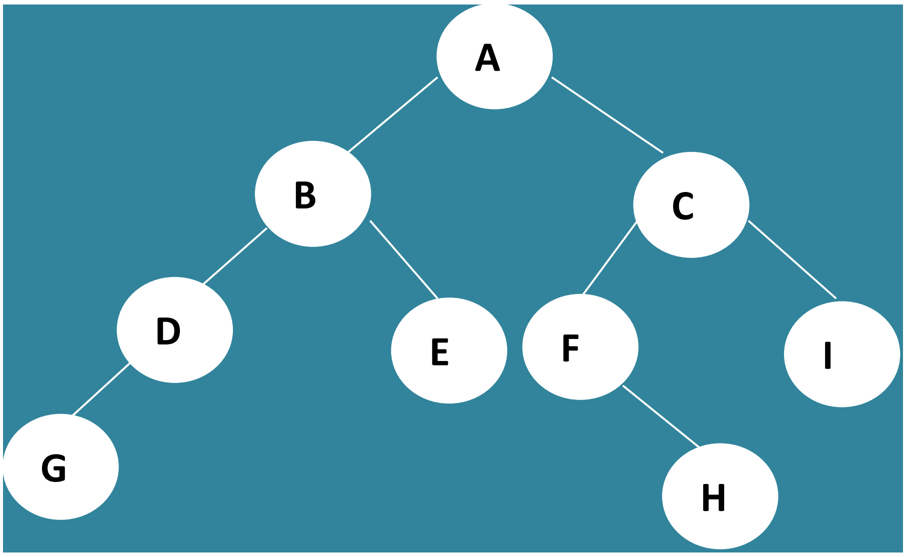
a) A B C * + D – E F + /



b) + * * - A B + C D E F

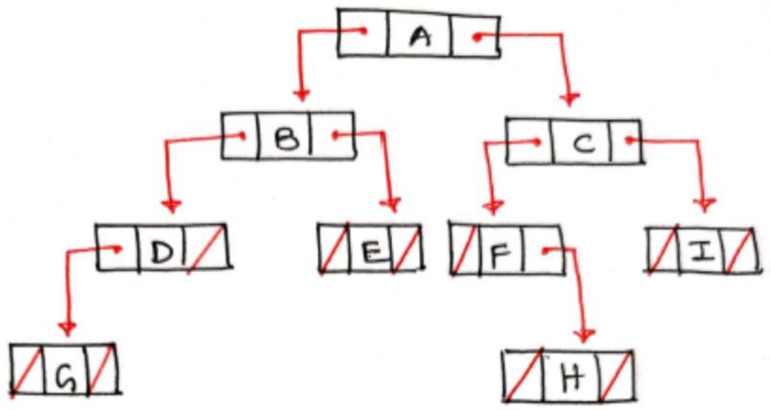


5. Give the array and Linked List representation for the tree structure



i	0	1	2	3	4	5	6	7	8	9	10	11	12	...	15
A[i]	A	B	C	D	E	F	I	G					H		

Array representation



Linked List Representation